Meridian: Toward a Computational Infrastructure for Cross-Domain Structural Analogy

Preliminary Research Note

ABSTRACT

Cross-domain structural analogy (the transfer of relational frameworks from one knowledge domain to another) is among the most historically reliable mechanisms underlying scientific and technological breakthroughs. Despite its demonstrated importance, no systematic computational infrastructure exists for its discovery. This paper introduces Meridian, a proposed system for identifying, representing, and generating cross-domain structural analogies through geometric operations over relational embedding spaces. The central hypothesis is that structurally parallel concept neighborhoods across distinct domains exhibit similar geometric configurations in embedding space, independent of surface vocabulary. We describe a prototype implementation, an experimental pipeline for hypothesis validation, a designed architecture comprising five principal components, and a set of open empirical questions. The primary technical contribution is the formulation of structural analogy as a shape-matching problem over relational vector spaces, and the identification of gap detection (concepts in one domain with no structural analog in another) as a generative discovery primitive. Key challenges include the inadequacy of general-purpose embeddings for structural similarity tasks and the need for a purpose-trained structural embedding model.

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1. INTRODUCTION

The history of scientific discovery exhibits a recurring pattern: transformative advances frequently originate not from the deepest specialists in a field but from researchers who import conceptual frameworks from adjacent or unrelated domains. Shannon (1948) derived the mathematical foundations of information theory from Boltzmann's thermodynamic entropy formula. Watson and Crick's elucidation of DNA structure was enabled by Crick's prior training in X-ray crystallography (Watson, 1968). The Wright brothers applied mechanical principles from bicycle engineering to the problem of powered flight. Kuhn (1962) documented this pattern systematically, observing that paradigm shifts characteristically originate at the periphery of established fields rather than from their most credentialed practitioners.

The common mechanism underlying these cases is structural analogy: the recognition that a relational framework developed in one domain to describe one class of phenomena applies, with appropriate reinterpretation of terms, to a structurally identical class of phenomena in another domain. This recognition is historically accidental, contingent on the biographical particulars of individual researchers. No systematic computational infrastructure exists to search for such parallels across the full space of scientific and technical literature.

This paper introduces Meridian, a system designed to provide that infrastructure. The central contribution is the formulation of cross-domain structural analogy as a *geometric shape-matching problem* over relational embedding spaces, and the proposal that gaps in this matching (concepts in one domain with no structural analog in another) constitute a generative discovery primitive: a mechanism for identifying candidate conceptual transfers before they have been articulated by any human researcher.

The paper is organized as follows. Section 2 reviews relevant prior work. Section 3 presents the core theoretical framework. Section 4 describes the current prototype and experimental pipeline. Section 5 details the designed system architecture. Section 6 addresses the central technical challenge of structural embeddings. Section 7 describes the mirror visualization interface and gap detection mechanism. Section 8 presents open empirical questions. Section 9 situates the work relative to existing approaches. Section 10 discusses current status and future directions.

2. RELATED WORK

2.1 Cognitive Theories of Analogy

The theoretical foundation for computational analogy research is Gentner's (1983) Structure-Mapping Theory, which proposes that analogical reasoning involves the alignment of relational structure across domains, with surface attribute similarity playing a secondary role. Gentner and Markman (1997) demonstrated empirically that productive analogies are characterized by systematic structural correspondence rather than object-level similarity. These findings motivate the present approach: if human analogical reasoning operates on relational structure, computational analogical retrieval should do the same.

2.2 Computational Analogy

Early computational approaches to analogy include the Structure-Mapping Engine (SME; Falkenhainer et al., 1989), which operationalized Gentner's theory as an algorithm for aligning predicate calculus representations of two domains. The Analogical Constraint Mapping Engine (ACME; Holyoak and Thagard, 1989) and MAC/FAC (Forbus et al., 1995) extended this work. These systems operate over hand-crafted symbolic representations and do not scale to arbitrary text corpora. The present work pursues a learned geometric approach that inherits the structural emphasis of SME while operating over dense vector representations derived automatically from text.

2.3 Geometric Properties of Embedding Spaces

Mikolov et al. (2013a) demonstrated that semantic relationships are encoded as consistent vector offsets in word embedding spaces, exemplified by the relation: vector(king) - vector(man) + vector(woman) is approximately vector(queen). This finding establishes that relational structure has geometric correlates in embedding spaces. Mikolov et al. (2013b) further showed that cross-lingual semantic alignment is achievable through linear transformation, providing a technical precedent for the cross-domain alignment operations proposed here.

2.4 Knowledge Graph Embeddings

Knowledge graph embedding methods, including TransE (Bordes et al., 2013), RotatE (Sun et al., 2019), and ComplEx (Trouillon et al., 2016), encode relational structure through geometric operations on entity and relation vectors. TransE models a relation r as a translation such that head + r is approximately tail, encoding the relational role geometrically. These methods are directly relevant to the structural embedding problem identified in Section 6.

2.5 AI for Scientific Discovery

Recent work has explored the use of large language models and retrieval systems to accelerate scientific discovery (Jumper et al., 2021; Taylor et al., 2022). Systems combining code execution with natural language retrieval have demonstrated early capability in hypothesis generation (Romera-Paredes et al., 2023). Meridian is complementary to this body of work: rather than accelerating search within a single domain, it provides infrastructure for importing conceptual frameworks across domain boundaries.

3. THEORETICAL FRAMEWORK

3.1 Analogy as Geometric Shape

We propose the following formal characterization. Let D_A and D_B denote two knowledge domains, each associated with a corpus C_A and C_B respectively. Let f: text -> R^d denote an embedding function mapping text to a d-dimensional vector space. A concept neighborhood N(c, D, k) is defined as the set of the k concepts in domain D most semantically proximate to concept c under f.

The geometric signature of a neighborhood N is its Gram matrix $G = VV^T$, where V is the matrix of mean-centered, scale-normalized neighborhood vectors. G encodes all pairwise dot products among neighborhood members, capturing both angular relationships and relative magnitudes while remaining invariant to global rotation and reflection.

Definition (Structural Analogy): Concepts c_A in D_A and c_B in D_B are structurally analogous with fidelity score s if:

$$s = \mathrm{sim}(G(N(c_A, D_A, k)), G(N(c_B, D_B, k))) > t$$

where sim denotes cosine similarity over flattened Gram matrices and t is an empirically determined threshold. This formulation is invariant to the absolute orientations of the embedding spaces, requiring no cross-space alignment.

3.2 Shape Projection

Given a concept neighborhood in domain A, the shape projection operation transfers its geometric configuration into domain B. Let mu_A denote the centroid of N(c_A, D_A, k) and {$delta_i$} the set of offset vectors from mu_A to each neighborhood member. The projection proceeds as follows:

(1) Identify anchor: c* = argmax over c in DB of sim(f(c), muA)

(2) Estimate local scale: sigmaB = percentile of {||f(c) - f(c*)||} at the 25th percentile, for c in NB(c*, k)

(3) Scale offsets: delta'i = deltai * (sigmaB / sigmaA)

(4) Project: p'i = f(c*) + delta'i

(5) Resolve: c'i = argmax over c in DB of sim(f(c), p'i)

The per-concept confidence of the projection is the cosine similarity between the projected point p'_i and the resolved concept $f(c'_i)$. Low confidence at a given position indicates that domain B lacks a concept in the structural role occupied by the corresponding domain A concept. These low-confidence positions constitute the gap set, described in Section 7.

3.3 The Inadequacy of Surface Similarity

The proposed approach is motivated by the failure of surface similarity to capture structural analogy. Consider the following illustrative cases. The expression "predator hunts prey" and the expression "editor cuts weak sentences" are syntactically parallel and share a general semantic structure (an agent acts to eliminate an object). Under general-purpose embedding similarity metrics, they score relatively highly. Yet they are not structurally analogous: the predator-prey relationship is embedded within a population-level selective system involving fitness landscapes, coevolution, and frequency-dependent dynamics, whereas the editor-sentence relationship is a quality filtering operation with none of these properties.

Conversely, "mutation introduces variation into a population" and "interest rate changes introduce volatility into markets" are structurally nearly identical, as both describe a stochastic perturbation mechanism generating diversity within a population subject to selective pressure. Yet they score lower under general embedding similarity due to vocabulary divergence. A system optimizing for surface similarity will preferentially return the former class of match while systematically missing the latter. The structural embedding problem is the challenge of building representations for which this ordering is reversed.

4. CURRENT PROTOTYPE AND EXPERIMENTAL PIPELINE

4.1 Language-Layer Prototype

A web-based prototype has been implemented that demonstrates the product concept. The system accepts a source domain, target domain, problem description, and analysis depth parameter from the user. It constructs a structured prompt specifying the analytical task (identifying non-obvious structural transfers, explaining the transferable mechanism, and identifying the precise point at which the analogy fails) and submits this to a large language model (Anthropic Claude Sonnet). The response is parsed as structured JSON and rendered with per-transfer fidelity and novelty scores.

This prototype establishes a baseline and serves as the interface layer for the full system. Its fundamental limitation is that it operates entirely within the language model's training distribution: the structural parallels it surfaces are those that have been articulated somewhere in the training corpus. It cannot, in principle, surface structural parallels between domains that have never been written about together. The geometric retrieval architecture described in subsequent sections is specifically designed to address this limitation.

4.2 Experimental Pipeline

A Jupyter notebook pipeline has been designed and implemented to test the core hypothesis: that geometric shape similarity in relational embedding space correlates with human-judged structural analogy quality at a rate significantly greater than chance. The pipeline comprises the following stages:

(i) Corpus acquisition via the Wikipedia API for selected domain topic lists;

(ii) Relational triple extraction using spaCy dependency parsing, extracting (subject, relation, object) triples with filtering for semantic content;

(iii) Triple embedding using OpenAI text-embedding-3-small, with the embedded representation being the concatenated triple string;

(iv) Cross-domain similarity search via matrix multiplication over normalized embedding vectors;

(v) Human evaluation using a three-point scoring protocol: 0 (surface coincidence), 1 (weak structural parallel), 2 (genuine structural analogy);

(vi) Language model evaluation for comparison with human scores, using a structured prompt requesting binary genuine/not-genuine judgment with confidence.

The primary outcome measure is the signal rate: the proportion of top-ranked geometric matches receiving a human score of 1 or 2. A signal rate significantly exceeding the rate achievable by random sampling from the cross-domain pair space would constitute evidence in support of the core hypothesis.

The pipeline has been designed but not yet fully executed. The initial experiment uses evolutionary biology and organizational management as the domain pair, with a secondary validation using immunology and cybersecurity, a pair selected for its known structural parallels, providing a positive control.

5. SYSTEM ARCHITECTURE

The full Meridian system comprises five principal components. The key architectural decision, that all cross-domain projections are precomputed offline and stored, has significant implications for system behavior, discussed in Section 5.2.

5.1 Component Overview

Component 1 (Domain Corpus and Concept Extraction): This component ingests text corpora for each domain, extracts relational triples via dependency parsing, deduplicates near-synonym concepts, and stores concept vectors in a per-domain namespace in a vector database. This component is the productionized form of the experimental pipeline described in Section 4.2.

Component 2 (Precompute Worker): An offline batch process that, for each domain pair (DA, DB), executes the shape projection algorithm for every concept in DA into DB and every concept in DB into DA, identifies high-confidence bridges and low-confidence gaps, and stores results in a relational database. This component runs once per domain pair and is re-executed only when the underlying corpus changes.

Component 3 (Idea Generation Layer): A component that accepts a gap concept (a concept in domain B with no structural analog in domain A) and its partial matches (domain A concepts proximate to the projected gap positions) and queries a large language model to articulate the gap concept expressed in domain A's vocabulary. The prompt specifies that the task is not analogical retrieval but conceptual translation: the construction of a genuinely new concept in domain A that did not previously exist.

Component 4 (API Layer): A FastAPI service exposing endpoints for domain management, precomputed projection retrieval, gap querying, idea generation, and user feedback collection.

Because the expensive computation is precomputed, the majority of endpoints are fast reads from the relational database.

Component 5 (Mirror Visualization Interface): A frontend application described in detail in Section 7.

5.2 The Precomputed Atlas

The decision to precompute all cross-domain projections offline, rather than computing them in response to user queries, transforms the fundamental character of the system. A query-time system is a retrieval engine in which users submit questions and receive answers. A precomputed system is a map in which the structural relationships between domains exist as a stable, queryable artifact independent of any particular user interaction.

This distinction has several important consequences. First, the gaps (concepts in one domain with no structural analog in the other) are visible as first-class objects in the map rather than being invisible absences in a retrieval system. Second, the atlas constitutes a scientific object that can be versioned and compared across time, potentially revealing how structural relationships between fields evolve as their literatures develop. Third, implicit feedback is collectible without explicit user rating: the portions of the atlas that users explore, the gaps they investigate, and the generated concepts they confirm or reject constitute behavioral signal that can improve the system without requiring active evaluation effort.

6. THE STRUCTURAL EMBEDDING PROBLEM

The most significant open technical problem in the Meridian architecture is the inadequacy of general-purpose embedding models for the structural similarity task. As demonstrated in Section 3.3, general embeddings trained on semantic similarity objectives conflate surface vocabulary similarity, syntactic similarity, and genuine structural similarity in ways that systematically disadvantage the retrieval of genuine structural analogs.

The required representation is one for which the distance metric captures relational role similarity independent of domain vocabulary. We term this a structural embedding. No such embedding currently exists as an off-the-shelf artifact.

The construction of a structural embedding model requires: (i) a training dataset of relational expression pairs labeled as structurally parallel or non-parallel; (ii) a contrastive training objective that minimizes distance between structurally parallel pairs while maximizing distance between structurally non-parallel pairs, including the difficult case of pairs that are surface-similar but structurally distinct; and (iii) a base model with sufficient representational capacity.

The experimental pipeline described in Section 4.2 serves a dual function: it provides validation data for the baseline geometric system and simultaneously generates the labeled pair dataset required to train the structural embedding model. Every human-scored pair (score 2 pairs as positive examples, score 0 pairs as hard negatives) constitutes a training instance. The experimental program is therefore also a data collection program.

Several intermediate approaches warrant investigation prior to full structural embedding development. Knowledge graph embedding models, specifically TransE (Bordes et al., 2013) and RotatE (Sun et al., 2019), encode relational structure geometrically through translation and rotation operations respectively and may capture structural similarity more faithfully than sentence embedding models. Fine-tuned sentence transformer models (Reimers and Gurevych, 2019) trained on the human-scored pair dataset represent a lower-resource alternative. The experimental comparison of these approaches against the general embedding baseline is among the open empirical questions in Section 8.

7. THE MIRROR INTERFACE AND GAP DETECTION

7.1 Interaction Model

The standard interaction model for retrieval systems is query-based: the user specifies what they are looking for and the system returns matching items. This model has a fundamental epistemological limitation in that the user can only retrieve what they know to ask about. A concept that does not yet exist in the user's domain cannot be queried.

The mirror interface addresses this limitation through an exploration-based interaction model. Two domains are displayed side by side, their concepts arranged spatially according to their geometric relationships in the embedding space, using UMAP dimensionality reduction to produce a two-dimensional layout that preserves local neighborhood structure. Structural connections between domains (concept pairs with fidelity scores exceeding threshold t) are rendered as lines crossing the central mirror axis, with line opacity encoding connection strength. Concepts with no high-confidence connection to the opposite domain are rendered as empty circles.

7.2 Gap Detection as Discovery Primitive

The empty circles, referred to as gap concepts, are the primary discovery mechanism of the system. A gap in domain A's projection into domain B indicates that domain B has developed a conceptual structure for which domain A has no equivalent. This is not a failure of the retrieval system; it is a substantive finding about the asymmetric development of conceptual frameworks across fields.

To illustrate: punctuated equilibrium, kin selection, the founder effect, and frequency-dependent selection are highly developed theoretical constructs in evolutionary biology describing dynamics that almost certainly operate in organizational systems. None has a formal equivalent in organizational management theory. These would appear as gap concepts in a projection of evolutionary biology onto organizational management, as they would land in the embedding space of organizational management without finding a close neighbor, indicating the structural position is occupied in biology but not in management.

When a user selects a gap concept, the idea generation layer constructs a description of what that concept would look like expressed in the target domain's vocabulary. This is not analogical retrieval. It is conceptual translation: the articulation of a structural framework that exists in one domain but has not yet been named or formalized in another. The output is a proposed new concept, a framework that did not exist in the target domain's literature prior to the interaction.

7.3 Three Operational Modes

The system supports three operational modes that build on one another sequentially:

Mode 1 (Structural Mapping): Displays existing structural connections between domains, answering the question of what relational knowledge the two fields share without knowing it.

Mode 2 (Gap Detection): Highlights absent connections, answering the question of what concepts in one domain have not transferred to the other.

Mode 3 (Concept Generation): Constructs articulations of gap concepts in the target domain's vocabulary, answering the question of what new conceptual framework the structural gap points toward.

This progression moves from retrieval (Mode 1) to identification (Mode 2) to generation (Mode 3), with each mode building on the outputs of the preceding one.

8. OPEN QUESTIONS

The following questions are empirically tractable and constitute the immediate research agenda:

Q1. Does geometric shape similarity between relational triple embeddings correlate with human-judged structural analogy quality at a rate significantly exceeding chance? What signal rate threshold constitutes meaningful evidence for the core hypothesis?

Q2. Do full-sentence embeddings produce superior cross-domain structural signal relative to triple-string embeddings? Does a hybrid representation outperform either alone?

Q3. Do knowledge graph embedding approaches (TransE, RotatE) produce superior structural signal relative to sentence embedding approaches on the cross-domain analogy retrieval task?

Q4. Does density-adaptive scaling in the shape projection operation improve projection accuracy across domain pairs with substantially different local concept densities?

Q5. Does fine-tuning a sentence transformer model on human-scored analogical pairs produce a structural embedding with meaningfully superior signal relative to the general embedding baseline? What is the minimum training set size required to observe improvement?

Q6. Do gap concepts identified by the system correspond to concepts that domain experts recognize as genuinely absent from the target domain, or does the gap detection mechanism primarily surface vocabulary incompatibilities and domain-specific technical terms with no meaningful structural equivalent?

Q7. Does the language model judgment of genuine/not-genuine correlate with human judgment sufficiently to serve as a scalable proxy for human evaluation in the training data collection pipeline?

9. RELATIONSHIP TO EXISTING APPROACHES

Meridian is distinguished from related systems along several dimensions. Retrieval-augmented generation (RAG) systems retrieve text passages semantically similar to a query and condition language model generation on the retrieved content. Meridian retrieves structural configurations rather than passages and generates conceptual frameworks rather than grounded answers. The retrieval primitive is fundamentally different.

Knowledge graph systems encode explicit relationships identified and entered by human curators. Meridian derives implicit relational structure from corpus text, enabling coverage of relationships that have not been explicitly named. The relational structure Meridian operates over is latent rather than curated.

Prompted language model analogy generation produces analogies drawn from the model's training distribution, specifically parallels that have been articulated, directly or indirectly, in the training corpus. Meridian's geometric retrieval is specifically designed to surface structural parallels between domains that have not been in prior conversation, representing a distinct and complementary capability.

Cross-lingual embedding alignment methods (Conneau et al., 2018) learn linear transformations that align embedding spaces for different languages, enabling cross-lingual retrieval. Meridian's shape-matching approach does not require such alignment, operating instead on within-space geometric signatures. This is both a technical distinction and a practical advantage: cross-lingual alignment requires known translation pairs as anchors, whereas Meridian's Gram matrix comparison requires no such cross-domain reference points.

10. CURRENT STATUS AND FUTURE DIRECTIONS

The current state of the system is as follows. A language-layer prototype demonstrating the product concept and interaction model has been implemented and is publicly available. An experimental pipeline for hypothesis validation has been designed and implemented but not yet executed. The system architecture has been specified at the component level. The theoretical framework, including the formal definition of structural analogy as Gram matrix similarity and the shape projection algorithm, has been developed.

The immediate priority is execution of the notebook experiment to obtain an initial signal rate estimate. This result determines the subsequent research direction: a signal rate substantially exceeding chance supports proceeding to shape projection implementation and precompute worker development, while a signal rate at or near chance indicates that the triple extraction or embedding approach requires revision before the geometric hypothesis can be meaningfully tested.

Following successful hypothesis validation, development priorities are: (i) implementation and evaluation of the shape projection algorithm against the baseline; (ii) construction of the precompute worker and domain pair atlas; (iii) development of the mirror visualization interface; (iv) execution of the structural embedding experiments (Q3 and Q5 above); and (v) implementation of the idea generation layer with evaluation against domain expert judgment.

The structural embedding model, a purpose-trained representation in which structurally parallel relational expressions from different domains are geometrically proximate, is the research contribution that does not yet exist. Its development is contingent on sufficient human-scored training pairs, which the experimental program generates as a byproduct. It represents the boundary between the engineering work of assembling existing components and the research work of producing something genuinely new.

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